

CIS 9340 Principles of Database Management Systems Fall 2024

Group1 Project 10

Ad-Tracking CRM Tool

CIS 9340 Principles of Database Management Systems

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Introduction

Considering the expectation of tracking the performance and effectiveness of marketing campaigns of organizations, we propose a project designed to facilitate the comprehensive monitoring of the entire marketing process. Before designing the project, our first step is to gather detailed requirements to ensure the solution meets client needs.

The main functions of our CRM tool include:

1. Quick Access to Information: The ability to view contract information at any time.
2. Efficiency in Problem Resolution: The tool allows for tracking advertising campaigns, with a ticket function to post issues about the advertisement.
3. Comprehensive System Tracking: The system offers full monitoring of the entire advertising process, including pre-launch, ongoing, and post-launch phases.
4. Data Security: The tool will employ encryption and regular maintenance to ensure client information security.
5. Information Accessibility: The system will be widely accessible, supporting simultaneous access by multiple users. Primary users include marketing managers from customer organizations, channel partners, customer support teams, and system administrators.

The innovation of this project lies in its interactive nature with clients and its real-time tracking of advertising effectiveness. Clients will be able to monitor their advertising campaigns' status in real time. In case of issues, they can initiate support tickets for quick resolution by the support team. This ensures an efficient, responsive approach to managing marketing campaigns.

The following outlines the steps we followed in designing our solution:

1. Conceptual Model using Lucid Chart: We utilized Lucid Chart to visualize and design the conceptual model, ensuring a clear understanding of business requirements and data relationships.
2. Logical Model using MySQL Workbench: Based on the conceptual model, we created the logical model in MySQL Workbench, defining the attributes of tables and identifying primary and foreign keys.
3. Physical Model using MySQL: The physical database was created using MySQL, and we generated data along with SQL scripts for inserting data through Mockaroo.
4. Data Implementation using Mockaroo: We used Mockaroo to generate mock data for populating the database.
5. CRM Tool using Retool: Finally, we used Retool to build a CRM system by integrating the data model and connecting it to the physical database.

What the Business do

Market Dynamics is an agency that helps organizations advertise their products and services through various channels, such as television, the Internet, radio, and supermarkets. Organizations, which are our customers, sign contracts with our company to specify the type of services or products, advertising channels, and costs. They send their advertisements to our company, and we place them in different channels according to the contract requirements. We also provide a platform for organizations to manage their advertising.

We are going to build a CRM tool to support data management to the organizations, including helping organizations to manage contracts, campaign Ad, and provide their services. The tool also features a ticket system, allowing organizations to create, monitor, and update tickets for various issues and tracking their statuses in real-time. Users can add

detailed comments and collaborate seamlessly to resolve problems. In addition to issue resolution, the CRM tool will enable organizations to analyze performance and track progress effectively. Through this tool, organizations can easily track the performance of their campaigns, ensuring they meet their goals effectively and make data-driven decisions that align with their strategies based on the analysis.

Functional Requirements

First, we need an information storage system to store the detailed information of customer organizations, including their name, products, and services. This system enable the company to maintain a centralize database of customer information that make it easier to obtain customer information and contract details when needed.

Secondly, we require a campaign and channel management that allow customers to select advertising channels (e.g., TV, radio, social media). And then our company can schedule campaigns based on contract terms and assign them to specific channels. It will also help both the company and its customers monitor campaign progress and evaluate performance, ensuring campaigns run efficiently.

Then, we need a ticketing system that organizations can create tickets for issues related to products, services, contracts, goods, or campaigns, and enable comments for communication between organizations and support staff. The ticketing system will display ticket status, such as open, in progress, resolved, and closed, allowing staff and organizations to track the progress of issues in real time.

Data Requirements

CRM tools require a structured database to manage various data entities. Key data requirements include capturing detailed information about organizations, such as their

contracts, along with specifics about products, services, goods, and advertising channels. The system should create a comprehensive ticketing system to track issues related to contracts and assets (including products, services, and goods). Each ticket should include descriptions, issue dates, status, comments, and staff to monitor the tickets. Additionally, the database must include service types, product categories, organization types, and advertising channels, ensuring relationships between these entities are clearly defined to support reporting, analytics, and operational workflows.

Regulatory Requirements

It is important to secure and back up the database. We should restrict access to confidential, private information, such as login credentials, organization information, and ticket details. Also, make sure the authorized person has access to the proper data. Moreover, make sure the campaign Ad is following the Anti-Spam and Marketing Regulations, for example, obtain consent from the users before sending Campaign Ad.

Costs

A powerful system will be continuously upgraded and optimized, so maintenance and development costs are inevitable. The primary costs of our project are as follows:

1. **System Development Costs**

Innovation is crucial for the system, as continuously improving its features will better meet customer needs and increase operational efficiency. The tool is currently in its initial development phase, and more features will be added in the future. Additionally, as the number of users increases, we will scale the system's capacity to support more clients simultaneously while ensuring robust encryption and data protection functions.

Detailed planning and execution during this stage will minimize future costs related to system failures or redesigns.

2. Maintenance and Support Costs

The tool requires regular maintenance to ensure it operates smoothly. These costs include periodic updates, bug fixes, and user support services. A well-structured support system will reduce downtime and help maintain user satisfaction.

3. Training Costs

To maximize the effectiveness of the new system, comprehensive training for both staff and customers will be necessary. Investment in training will ensure that all users are proficient in using the system's functionalities. Training costs are vital for realizing the full potential of the system and achieving operational efficiency.

Benefits

1. Effectively Monitor and Manage Issues

One of the primary benefits of the tool is its ability to effectively monitor and manage relevant issues. By opening tickets in the system, customers can provide feedback at any time when they encounter a problem.

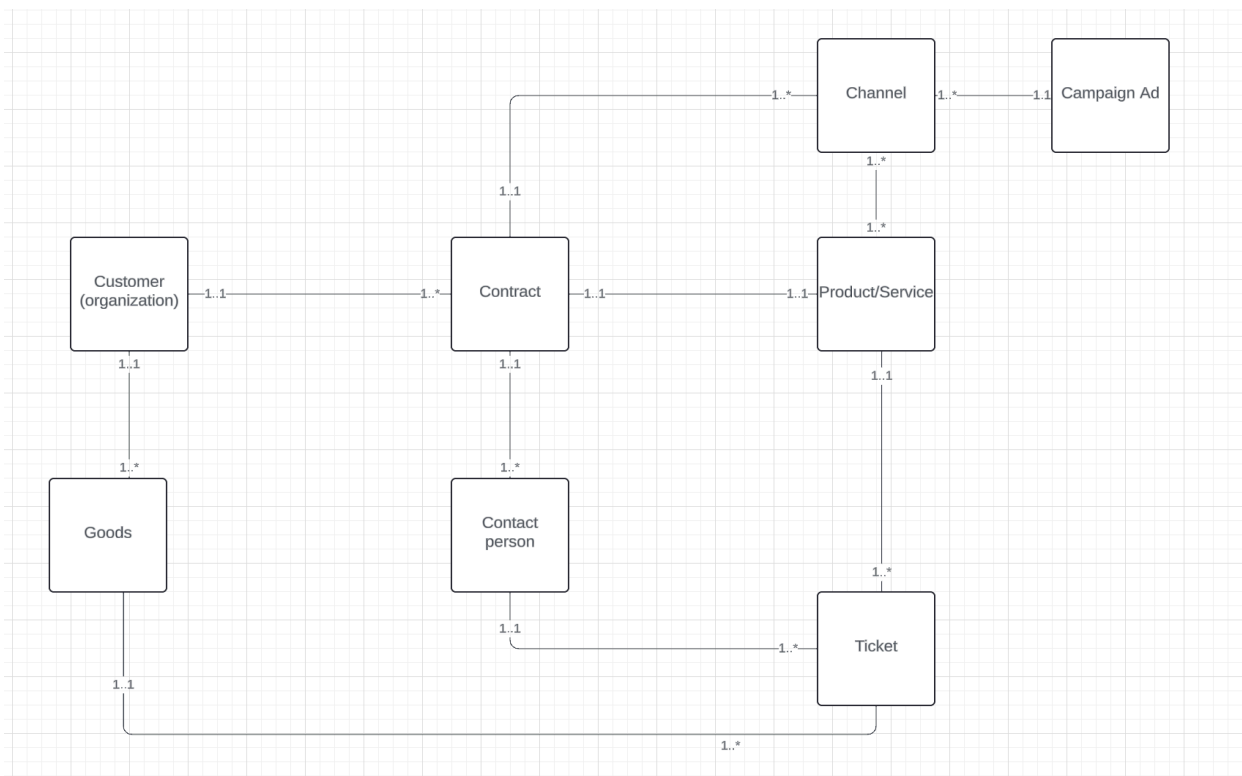
2. Quick Response and Resolution

Once customers open a ticket to report an issue, staff can quickly assess and resolve the problem. This not only enhances operational efficiency but also reduces the likelihood of risks and losses for organizations. With automated alerts and communication channels, the response time to issues will be significantly reduced. This benefit is crucial for maintaining service quality and customer satisfaction, as it ensures operational hiccups are addressed swiftly.

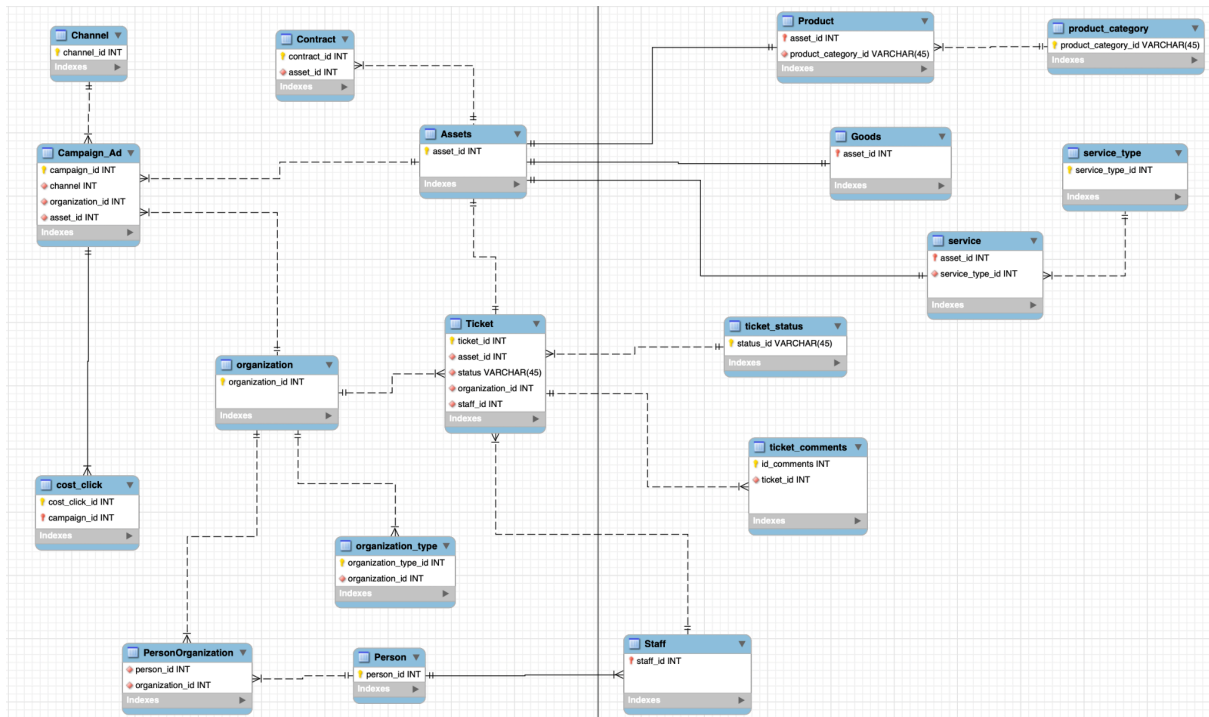
3. Real-Time Reporting and Analytics

Reporting and analytics capabilities will be provided to organizations to visualize the performance of advertisements and user behavior. This updated data is helpful for decision-making and strategic planning, enabling organizations to adapt to changing market conditions and meet customers' needs effectively.

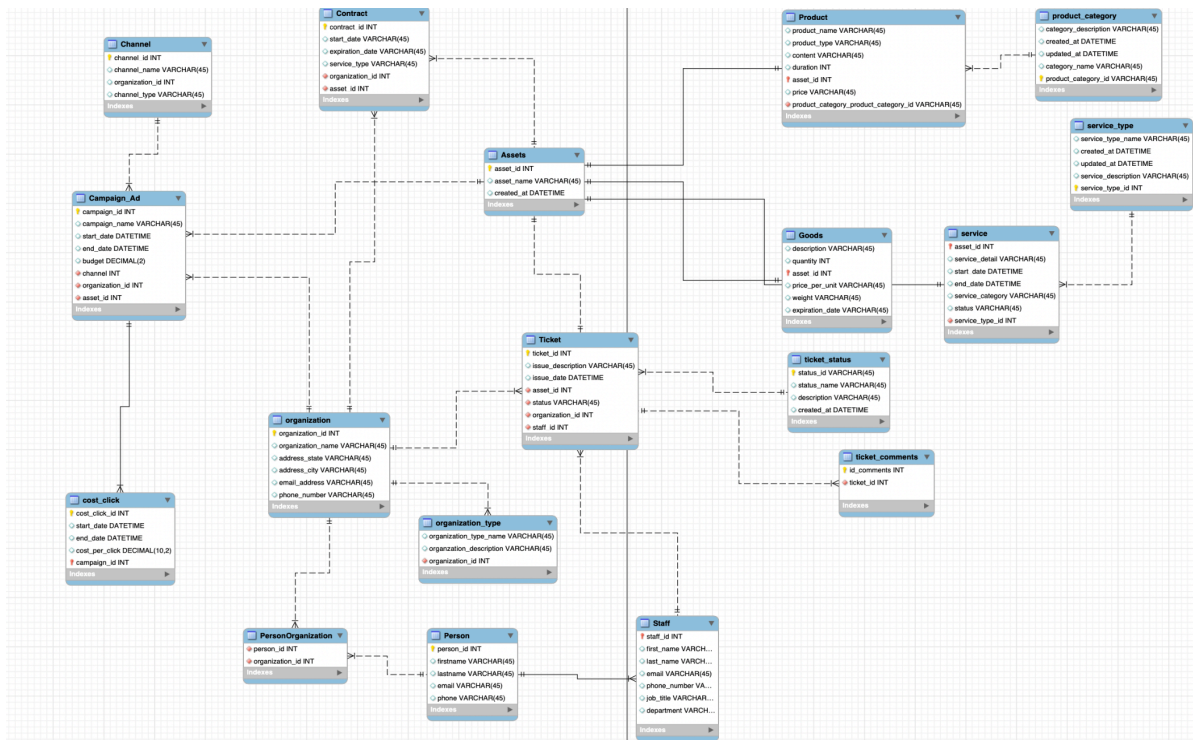
Conceptual Diagram



Logical Diagram



Physical Diagram



Normalized Relations and Functional Dependencies

The database for Market Dynamics was designed to reduce redundancy, ensure data integrity, and organize information efficiently. To achieve this, we used normalization techniques, specifically up to Third Normal Form (3NF). This process included identifying functional dependencies, decomposing tables, and resolving redundancy issues step by step.

We began by achieving First Normal Form (1NF). In this step, we made sure that all columns contained only atomic values, meaning no repeating groups or multiple values in a single column. For example, in the Channel table, channel types like "TV," "Radio," and "Internet" were stored as individual rows instead of being combined in one column.

Next, we moved to Second Normal Form (2NF), which focused on eliminating partial dependencies. Partial dependency happens when a non-key column depends only on part of a composite primary key. To fix this, we created separate tables where all non-key attributes fully depended on the primary key. For instance, the Contract table included details like Cost and Duration, and these attributes depended directly on the ContractID.

Finally, we ensured Third Normal Form (3NF) by removing transitive dependencies. Transitive dependency occurs when a non-key attribute depends on another non-key attribute. For example, in the Customer table, CustomerName and ContactInfo were directly dependent on CustomerID, not on other attributes. Similarly, in the Goods table, attributes like Description, Quantity, and Weight were fully dependent on GoodsID. We identified functional dependencies for each table. Examples include:

In the Customer table: $\text{CustomerID} \rightarrow \text{CustomerName}, \text{ContactInfo}$

In the Contract table: $\text{ContractID} \rightarrow \text{CustomerID}, \text{ChannelID}, \text{ProductID}, \text{Cost}, \text{Duration}$

In the Ticket table: $\text{TicketID} \rightarrow \text{ContractID}, \text{ProductID}, \text{GoodsID}, \text{Status}, \text{Description}$

These dependencies ensured that all data was uniquely identified by a primary key. While normalization reduces redundancy and improves data consistency, it sometimes requires multiple table joins, which can affect performance. In this project, we did not perform **denormalization** because the system's performance met the requirements, and query complexity was manageable. However, in real-world systems, some derived values like **TotalCost** or **TotalContracts** could be stored to improve query speed.

In conclusion, we achieved **3NF** by following clear steps: eliminating repeating groups (1NF), removing partial dependencies (2NF), and resolving transitive dependencies (3NF). This resulted in an organized, clean, and efficient database design that supports the CRM system's goals.

An Example of the SQL DDL

SQL Data Definition Language (DDL) statements are used to define database structures and relationships. To ensure data consistency, primary keys and foreign keys are implemented using methods such as CREATE TABLE to connect tables within the database, enabling seamless interaction between them.

Create the 'organization' table

```
CREATE TABLE IF NOT EXISTS `organization` (  
  `organization_id` INT NOT NULL,  
  `organization_name` VARCHAR(45) NOT NULL,  
  `address_state` VARCHAR(45),  
  `address_city` VARCHAR(45),  
  `email_address` VARCHAR(45),  
  `phone_number` VARCHAR(45),  
  PRIMARY KEY (`organization_id`)  
) ENGINE=InnoDB  
DEFAULT CHARSET=utf8;
```

Create the 'channel' table

```

CREATE TABLE IF NOT EXISTS `channel` (
  `channel_id` INT NOT NULL,
  `channel_name` VARCHAR(45) NOT NULL,
  `channel_type` VARCHAR(45),
  `organization_id` INT NOT NULL,
  PRIMARY KEY (`channel_id`),
  INDEX `fk_channel_organization_idx` (`organization_id` ASC),
  CONSTRAINT `fk_channel_organization`
    FOREIGN KEY (`organization_id`)
    REFERENCES `organization` (`organization_id`)
    ON DELETE CASCADE
    ON UPDATE CASCADE
) ENGINE=InnoDB
DEFAULT CHARSET=utf8;

```

Create the 'assets' table

```

CREATE TABLE IF NOT EXISTS `assets` (
  `asset_id` INT NOT NULL,
  `asset_name` VARCHAR(45),
  `created_at` DATETIME,
  PRIMARY KEY (`asset_id`)
) ENGINE=InnoDB
DEFAULT CHARSET=utf8;

```

Create the 'campaign_ad' table

```

CREATE TABLE IF NOT EXISTS `campaign_ad` (
  `campaign_id` INT NOT NULL,
  `campaign_name` VARCHAR(45),
  `start_date` DATETIME,
  `end_date` DATETIME,
  `budget` DECIMAL(10,2),
  `channel_id` INT NOT NULL,
  `organization_id` INT NOT NULL,
  `asset_id` INT,
  PRIMARY KEY (`campaign_id`),
  INDEX `fk_campaign_channel_idx` (`channel_id` ASC),
  INDEX `fk_campaign_organization_idx` (`organization_id` ASC),
  INDEX `fk_campaign_asset_idx` (`asset_id` ASC),
  CONSTRAINT `fk_campaign_channel`
    FOREIGN KEY (`channel_id`)

```

Create the 'ticket' table

```
CREATE TABLE IF NOT EXISTS `ticket` (  
  `ticket_id` INT NOT NULL,  
  `issue_description` VARCHAR(45),  
  `issue_date` DATETIME,  
  `asset_id` INT NOT NULL,  
  `organization_id` INT NOT NULL,  
  `staff_id` INT NOT NULL,  
  PRIMARY KEY (`ticket_id`),  
  INDEX `fk_ticket_asset_idx` (`asset_id` ASC),  
  INDEX `fk_ticket_organization_idx` (`organization_id` ASC),  
  INDEX `fk_ticket_staff_idx` (`staff_id` ASC),  
  CONSTRAINT `fk_ticket_asset`  
    FOREIGN KEY (`asset_id`)  
    REFERENCES `assets` (`asset_id`)  
    ON DELETE NO ACTION  
    ON UPDATE NO ACTION,  
  CONSTRAINT `fk_ticket_organization`  
    FOREIGN KEY (`organization_id`)  
    REFERENCES `organization` (`organization_id`)  
    ON DELETE NO ACTION  
    ON UPDATE NO ACTION,  
  CONSTRAINT `fk_ticket_staff`  
    FOREIGN KEY (`staff_id`)  
    REFERENCES `staff` (`staff_id`)  
    ON DELETE NO ACTION  
    ON UPDATE NO ACTION  
) ENGINE=InnoDB  
DEFAULT CHARSET=utf8;
```

Insert data into 'organization'

```
INSERT INTO `organization` (`organization_id`, `organization_name`,  
  `address_state`, `address_city`, `email_address`, `phone_number`)  
VALUES (1, 'Bruen and Sons', 'Alabama', 'New York', 'null',  
  '830-150-5179');
```

Insert data into 'channel'

```
INSERT INTO `channel` (`channel_id`, `channel_name`, `channel_type`,  
  `organization_id`)  
VALUES (10, 'Instagram', '10', TV);
```

Insert data into 'assets'

```
INSERT INTO `assets` (`asset_id`, `asset_name`, `created_at`)  
VALUES (23, 'books', '2024-02-28 10:20:00');
```

Insert data into 'campaign_ad'

```
INSERT INTO `campaign_ad` (`campaign_id`, `campaign_name`, `start_date`,  
`end_date`, `budget`, `channel_id`, `organization_id`, `asset_id`)  
VALUES (15, 'Holiday Specials', '2024-01-27 00:00:00', '2024-09-14  
00:00:00', 504.61, 15, 15, 15);
```

Insert data into 'ticket'

```
INSERT INTO `ticket` (`ticket_id`, `issue_description`,  
`issue_date`, `asset_id`, `organization_id`, `staff_id`)  
VALUES (2000, 'comment 1:Low Click-Through Rate (CTR)',  
'2024-08-24 00:00:00', 1, 1, 1);
```

Examples Printout of Each of the Forms

Dashboard Overview:

The dashboard provides overview of the CRM system include contracts, services type, and organizations. There is a form shows the amount of active contracts. Our company can use this form to manage contracts and check active contracts easily.

The first query retrieves all records from the contract table, and the second query counts the number of active campaigns in the campaign_ad table to create active contracts form.

```
SELECT * FROM contract;  
SELECT COUNT(*) AS active_campaigns  
FROM campaign_ad  
WHERE end_date > NOW();
```

Form title

Contract ID

1

Start date

Nov 15, 2024 12:00 AM

Expiration date

Nov 30, 2024 12:00 AM

Service type

Select an option

Organization ID

32

Asset ID

1

Submit

Contract ID

Search in table

Active Contracts

0

▲ 8%

Retool

Contract ID	Start date	Expiration date	Asset ID	Service type	Organization ID
1	Nov 15, 2024 12:00 ...	Nov 30, 2024 12:00 ...	1	Online	32
2	Apr 22, 2024 12:00 ...	Feb 3, 2024 12:00 AM	2	Online	26
3	Aug 31, 2024 12:00 ...	Oct 15, 2024 12:00 ...	3	Online	13
4	Mar 21, 2024 12:00 ...	Aug 1, 2024 12:00 AM	4	Online	38
5	Nov 26, 2024 12:00 ...	May 8, 2024 12:00 AM	5	Radio	8
6	Nov 17, 2024 12:00 ...	Apr 24, 2024 12:00 ...	6	Radio	10
7	Aug 26, 2024 12:00 ...	Aug 11, 2024 12:00 ...	7	TV	28
8	Nov 13, 2024 12:00 ...	Nov 2, 2024 12:00 AM	8	Radio	13
9	Nov 2, 2024 12:00 AM	Feb 18, 2024 12:00 ...	9	Online	7
10	Feb 10, 2024 12:00 ...	Jan 29, 2024 12:00 ...	10	Online	47
11	Jul 18, 2024 12:00 AM	Mar 10, 2024 12:00 ...	11	Online	32
12	Mar 29, 2024 12:00 ...	Jun 1, 2024 12:00 AM	12	Radio	35
13	Mar 6, 2024 12:00 AM	Feb 25, 2024 12:00 ...	13	Radio	49
14	Mar 11, 2024 12:00 ...	May 30, 2024 12:00 ...	14	Radio	41
15	Jan 23, 2024 12:00 ...	Jan 20, 2024 12:00 ...	15	Supermarket	6
16	Aug 7, 2024 12:00 AM	Nov 11, 2024 12:00 ...	16	Supermarket	23
17	Sep 2, 2024 12:00 AM	Jan 6, 2024 12:00 AM	17	Radio	12
18	Mar 10, 2024 12:00 ...	Mar 17, 2024 12:00 ...	18	TV	
19	Jan 14, 2024 12:00 ...	Feb 22, 2024 12:00 ...	19	Supermarket	

Good Management:

This form can use to manage goods sent by customers for advertising. It includes goods information, such as weight, height, quantity, and descriptions. This query retrieves all records from the goods table for goods associated with an organization name.

```
SELECT g.*
FROM goods g
INNER JOIN organization o ON g.asset_id = o.organization_id
WHERE o.organization_name = {{select3.value}}
OR {{select3.value}} IS NULL;
```

Asset ID	Description	Quantity	Price per unit	Weight	Expiration date
1	Item 1 Clothing	189	17.35	164.56	Jan 1, 2024
2	Item 3 Toys	204	3.98	110.23	Feb 1, 2024
3	Item 3 Toys	414	19.96	160.75	Mar 1, 2024
4	Item 3 Toys	466	9.46	268.03	Apr 1, 2024
5	Item 2 Food	451	2.52	239.62	May 1, 2024
6	Item 3 Toys	350	2.67	253.3	Jun 1, 2024
7	Item 1 Clothing	378	9.2	199.51	Jul 1, 2024
8	Item 4 Books	495	16.58	271.45	Aug 1, 2024
9	Item 2 Food	302	7.78	144.06	Sep 1, 2024

50 results

Campaign Management

This form shows the details of campaign, channels, services type, and organization information, which allows our company and organizations to create and manage campaigns. It also provides budget for each campaign.

This query retrieves detailed information about advertising campaigns, and filters campaigns started within the last year (optional) and sorting them in descending order of their start dates.

```
SELECT
c.campaign_id,c.campaign_name,c.start_date,c.end_date,c.budget,ch.
channel_name,ch.channel_type,o.organization_name,
    ct.start_date AS contract_start_date,
    ct.expiration_date AS contract_expiration_date,
    ct.service_type
FROM
    Campaign_Ad c
JOIN
    Channel ch ON c.channel = ch.channel_id
JOIN
    Organization o ON c.organization_id = o.organization_id
JOIN
    Contract ct ON ct.organization_id = o.organization_id
WHERE
    c.start_date >= NOW() - INTERVAL 1 YEAR -- Optional: Filter
campaigns from the last year
ORDER BY
    c.start_date DESC;
```

Campaign ID	Campaign name	Start date	End date	Budget	Channel name	Channel type	Organizatio...	Contract start date	Contract expiration date	Service type
4	Summer Campaign	Nov 21, 2024 12:00 AM	May 8, 2024 12:00 AM	321.96	TV	TV	Emmerich, Tromp and...	Aug 18, 2024 12:00 AM	May 21, 2024 12:00 AM	TV
4	Summer Campaign	Nov 21, 2024 12:00 AM	May 8, 2024 12:00 AM	321.96	TV	TV	Emmerich, Tromp and...	Aug 24, 2024 12:00 AM	Aug 13, 2024 12:00 AM	Supermarket
4	Summer Campaign	Nov 21, 2024 12:00 AM	May 8, 2024 12:00 AM	321.96	TV	TV	Emmerich, Tromp and...	Aug 16, 2024 12:00 AM	Feb 24, 2024 12:00 AM	Radio
19	Summer Campaign	Nov 13, 2024 12:00 AM	Apr 26, 2024 12:00 AM	853.56	You Tube	TV	Schneider, Stokes and...	Jul 22, 2024 12:00 AM	Oct 28, 2024 12:00 AM	Radio
30	Back To School	Oct 17, 2024 12:00 AM	Nov 20, 2024 12:00 AM	265.53	Instagram	TV	Conn-Watsica	Jun 25, 2024 12:00 AM	Oct 28, 2024 12:00 AM	Online
30	Back To School	Oct 17, 2024 12:00 AM	Nov 20, 2024 12:00 AM	265.53	Instagram	TV	Conn-Watsica	Sep 20, 2024 12:00 AM	May 8, 2024 12:00 AM	Radio
3	Holiday Specials	Sep 20, 2024 12:00 AM	Apr 28, 2024 12:00 AM	645.05	TV	Online	Fell-Fritsch	Nov 13, 2024 12:00 AM	Sep 15, 2024 12:00 AM	TV

50 results

Ticket Management

Ticket management forms include the details of organizations, issues, comments, assets, as well as staff information. Organizations and staffs can utilize this form to monitor tickets. This SQL query retrieves detailed information about tickets, by joining the ticket table with ticket_status, ticket_comments, assets, organization, and staff tables.

```
SELECT t.ticket_id AS ticket, t.issue_description AS issue,
t.issue_date,
    ts.status_name AS status,
    tc.comments,
    a.asset_name AS assets,
    o.organization_name AS organization,
    s.first_name AS FirstName, s.last_name AS LastName, s.email
FROM ticket AS t
INNER JOIN ticket_status AS ts ON t.status = ts.status_id
INNER JOIN ticket_comments AS tc ON t.ticket_id = tc.ticket_id
INNER JOIN assets AS a ON t.asset_id = a.asset_id
INNER JOIN organization AS o ON t.organization_id =
o.organization_id
INNER JOIN staff AS s On t.staff_id = s.staff_id;
```

Ticket	Issue	Issue date	Status	Comments	Assets	Organization	First name	Last name	Email
1	comment 1:Low Click-Through ...	Aug 24, 2024 12:00 AM	Closed	Good 4 5	Clothing	Bruen and Sons	Gerti	Aggus	gaggus0@canalblog.com
1	comment 1:Low Click-Through ...	Aug 24, 2024 12:00 AM	Closed	Excellent 5 5	Clothing	Bruen and Sons	Gerti	Aggus	gaggus0@canalblog.com
1	comment 1:Low Click-Through ...	Aug 24, 2024 12:00 AM	Closed	Average 3 5	Clothing	Bruen and Sons	Gerti	Aggus	gaggus0@canalblog.com
1	comment 1:Low Click-Through ...	Aug 24, 2024 12:00 AM	Closed	Excellent 5 5	Clothing	Bruen and Sons	Gerti	Aggus	gaggus0@canalblog.com
2	comment 1:Low Click-Through ...	Apr 29, 2024 12:00 AM	In Process	Excellent 5 5	Toys	Parisian-Ortiz	Garfield	Mongenot	gmongenot1@lycos.com
2	comment 1:Low Click-Through ...	Apr 29, 2024 12:00 AM	In Process	Fair 2 5	Toys	Parisian-Ortiz	Garfield	Mongenot	gmongenot1@lycos.com
2	comment 1:Low Click-Through ...	Apr 29, 2024 12:00 AM	In Process	Average 3 5	Toys	Parisian-Ortiz	Garfield	Mongenot	gmongenot1@lycos.com
2	comment 1:Low Click-Through ...	Apr 29, 2024 12:00 AM	In Process	Good 4 5	Toys	Parisian-Ortiz	Garfield	Mongenot	gmongenot1@lycos.com
3	comment 4: Technical error	Nov 15, 2024 12:00 AM	Resolved	Excellent 5 5	Clothing	Feil-Fritsch	Marcy	Demogeot	mdemogeot2@timesonline.co.u
3	comment 4: Technical error	Nov 15, 2024 12:00 AM	Resolved	Average 3 5	Clothing	Feil-Fritsch	Marcy	Demogeot	mdemogeot2@timesonline.co.u
3	comment 4: Technical error	Nov 15, 2024 12:00 AM	Resolved	Fair 2 5	Clothing	Feil-Fritsch	Marcy	Demogeot	mdemogeot2@timesonline.co.u
3	comment 4: Technical error	Nov 15, 2024 12:00 AM	Resolved	Poor 1 5	Clothing	Feil-Fritsch	Marcy	Demogeot	mdemogeot2@timesonline.co.u
4	comment 2: Ad Fatigue	Oct 8, 2024 12:00 AM	Awaiting	Fair 2 5	Toys	Emmerich, Tromp and Thiel	Marius	Liven	mliven3@nhs.uk

50 results



A Picture of the Navigation Form

CRM

PreviewingLatest

ShareEdit app

Logo

Dashboard

Good

Campaign_management

Ticket_management

Form title

Contract ID

1

Start date

Nov 15, 2024 12:00 AM

Expiration date

Nov 30, 2024 12:00 AM

Service type

Select an option

Organization ID

32

Asset ID

1

Submit

Contract ID

Search in table

Active Contracts

8%

Retool

Gross volume

Contract ID	Start date	Expiration date	Asset ID	Service type	Organization ID
1	Nov 15, 2024 12:00 ...	Nov 30, 2024 12:00 ...	1	Online	32
2	Apr 22, 2024 12:00 ...	Feb 3, 2024 12:00 AM	2	Online	26
3	Aug 31, 2024 12:00 ...	Oct 15, 2024 12:00 ...	3	Online	13
4	Mar 21, 2024 12:00 ...	Aug 1, 2024 12:00 AM	4	Online	38
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16	Aug 7, 2024 12:00 AM	Nov 11, 2024 12:00 ...	16	Supermarket	23
17	Sep 2, 2024 12:00 AM	Jan 6, 2024 12:00 AM	17	Radio	12
18	Mar 10, 2024 12:00 ...	Mar 17, 2024 12:00 ...	18	TV	34

production

main

Latest

Multipage Feedback

No queries running

Debug

Narrative Conclusion

Working on this project as a group was a valuable and challenging experience. Each step had its own difficulties, but we managed to work together to complete the project. The most difficult part was the physical model. When we started working on this step, one of our group members unexpectedly left the team. This caused a lot of problems because some of the work had to be redone. The logical part, which was already completed, needed to be revised and started over to make sure everything matched. At that point, we were also under pressure because the due date was very close. Balancing the workload, fixing errors, and ensuring the physical model was correct while managing time was very stressful for the team. The easiest part of the project was the conceptual model. This step required us to focus on the main idea of our system and carefully follow the project instructions. We did not need to write any complicated code on this step. Instead, we concentrated on identifying the core components of the system, such as customers, contracts, goods, and tickets, and defining how they connect to each other. By reading the project instructions carefully, we were able to determine the relationships between entities and build a clear and logical diagram. Since this

part focused more on planning and understanding the problem rather than the technical part, it was straightforward to complete.

Throughout this team project, we learned many valuable lessons. One key takeaway was understanding that unexpected challenges can happen during group work, and it is important to adapt and stay focused to overcome them. We also gained a deeper understanding of how databases work in real-world applications, including how data is structured, normalized, and connected to a system.

Even though the project had many parts to complete and required us to learn new concepts quickly, the experience was truly worth it. It allowed us to see how all the pieces of a database come together to build a functional system. Additionally, we explored tools like Retool and learned to develop applications with our own hands. This hands-on experience was not only educational but also fun. In the end, we realized the value of working hard on school projects because it prepares us for work in the real world.

If we had to do this project again, we would focus on creating a stronger plan at the beginning. This includes assigning clear roles to team members and avoiding falling behind. We would also allow extra time for unexpected problems, such as redoing parts of the project or dealing with technical errors. We believe that the proposed benefits of the new system can be realized. The system will make it easier for Market Dynamics to manage contracts, goods, and ticket issues. With clear relationships between tables, retrieving information like contract details, ticket comments, or goods promotion status will be much faster and more organized. The ticketing system will also improve communication by tracking issues and responses between the organization and the customer representative.

In conclusion, this project gave us a deeper understanding of database design and teamwork. Although the physical model was the hardest part due to unexpected challenges, we learned how to overcome problems under pressure.

